

SHERIDAN, WY, USA

WMO: 726660

Lat: 44.769N

Lon: 106.969W

Elev: 3945

StdP: 12.72

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24029

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.3	-4.0	-16.7	2.6	-9.8	-10.2	3.7	-3.6	32.6	30.7	28.4	31.6	3.5	310	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.0	95.3	63.4	92.0	62.9	88.7	62.2	67.6	86.2	65.8	84.6	64.2	83.1	7.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	94.1	75.1	59.4	87.5	73.1	57.3	81.2	71.7	34.1	86.1	32.7	84.3	31.3	83.0	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	28.0	24.3	20.1	DB	-19.4	100.9	5.9	2.7	-23.6	102.9	-27.1	104.5	-30.4	106.0	-34.7	108.0
(5)				WB	-19.5	71.4	5.4	2.6	-23.4	73.3	-26.5	74.8	-29.6	76.3	-33.5	78.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	24.4	26.0	35.6	43.5	52.3	62.1	71.5	69.5	59.4	45.8	33.9	24.4
	DBStd	19.21	12.40	12.88	11.56	8.96	7.82	7.12	5.39	5.86	8.58	9.50	11.63	12.16
	HDD50	3716	795	674	454	228	67	6	0	0	21	188	490	793
	HDD65	7470	1260	1093	910	646	398	134	12	27	204	596	933	1257
	CDD50	2189	0	1	9	32	138	369	666	605	304	58	7	0
	CDD65	468	0	0	0	0	3	47	213	167	37	1	0	0
	CDH74	8057	0	0	3	32	170	940	3232	2666	925	87	2	0
	CDH80	3714	0	0	0	2	33	352	1636	1302	368	21	0	0
Wind	WSAvg	6.9	6.5	7.1	7.7	8.9	8.2	6.8	6.1	5.9	6.1	6.7	6.3	6.4
Precipitation	PrecAvg	14.6	0.6	0.6	0.9	1.7	2.7	2.2	1.2	0.7	1.3	1.5	0.7	0.5
	PrecMax	19.4	1.3	1.3	1.9	3.5	6.3	4.1	3.8	2.3	4.3	3.7	2.1	1.1
	PrecMin	9.8	0.0	0.0	0.2	0.6	0.3	0.6	0.1	0.0	0.0	0.2	0.1	0.2
	PrecStd	2.5	0.4	0.4	0.5	0.7	1.7	0.9	0.9	0.5	1.1	0.8	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	62.5	72.5	78.7	84.0	94.4	100.4	98.8	93.5	83.1	70.5	57.8
		MCWB	42.2	44.7	49.5	54.0	60.2	63.0	64.5	64.2	61.1	56.1	49.1	42.1
	2%	DB	50.0	54.4	65.3	72.4	78.6	88.5	96.1	94.8	89.0	75.2	63.1	50.3
		MCWB	38.9	40.6	46.1	51.5	58.0	63.4	64.0	63.0	59.1	53.0	45.9	38.9
	5%	DB	45.1	48.0	60.0	66.9	74.3	83.7	92.7	91.3	84.2	69.0	57.4	45.5
		MCWB	36.3	37.5	44.3	49.4	56.9	62.6	63.6	62.6	58.2	50.9	43.3	36.2
	10%	DB	40.8	43.1	54.5	61.6	69.9	79.3	89.2	87.8	79.3	62.9	51.2	40.6
		MCWB	34.0	34.9	41.9	46.9	54.6	61.3	63.2	61.8	56.8	48.7	40.9	33.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.1	45.7	50.4	55.8	63.5	69.0	69.9	70.8	65.1	57.4	50.1	43.0
		MCDB	56.1	61.6	68.8	74.4	77.9	84.0	88.7	91.1	83.5	79.0	68.5	56.4
	2%	WB	39.3	41.3	47.3	52.6	60.1	66.2	67.8	66.6	61.9	54.6	46.7	39.4
		MCDB	49.2	53.4	64.0	69.1	74.7	81.9	87.1	86.9	81.8	72.2	61.6	49.5
	5%	WB	36.6	38.2	44.9	50.1	57.8	64.2	66.2	64.5	59.8	51.7	43.9	36.8
		MCDB	44.1	47.4	59.0	65.7	72.1	80.2	85.6	84.8	79.6	66.2	56.4	44.8
	10%	WB	34.4	35.3	42.2	47.7	55.5	62.2	64.7	62.8	57.6	49.3	41.1	33.9
		MCDB	40.3	42.0	53.5	60.4	68.7	77.8	84.3	82.9	76.4	62.5	50.9	39.9
Mean Daily Temperature Range	5% DB	MDBR	21.7	21.8	24.0	24.5	25.1	27.6	32.0	32.3	30.4	26.3	24.4	21.8
		MCDBR	26.1	28.3	34.2	35.6	34.4	36.4	38.3	39.0	39.7	37.5	33.6	27.0
		MCWBR	16.8	17.3	18.3	18.7	17.8	16.7	14.0	15.0	16.2	19.0	19.0	17.3
	5% WB	MCDBR	25.0	26.8	32.0	33.2	31.5	32.4	33.1	33.4	35.6	34.0	31.9	25.8
		MCWBR	16.6	16.7	17.7	17.8	17.2	16.6	13.9	14.8	15.3	18.0	18.2	17.1
Clear-Sky Solar Irradiance	taub	0.251	0.260	0.285	0.308	0.336	0.342	0.364	0.362	0.326	0.297	0.269	0.251	
	taud	2.534	2.515	2.493	2.446	2.434	2.445	2.415	2.396	2.488	2.536	2.529	2.527	
	Ebn at Noon	285	301	305	303	296	293	285	282	285	282	273	273	
	Edh at Noon	20	25	29	34	35	35	36	35	30	24	20	18	
All-Sky Solar Radiation	RadAvg	576	879	1260	1578	1858	2220	2292	1946	1510	967	634	474	
	RadStd	34	44	85	89	188	160	88	84	109	81	52	29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	-0.99	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page